Disease Prediction System Using Machine Learning

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TABLE OF CONTENTS

1. Project Description

MedAssist AI is a machine learning–based healthcare assistant designed to predict diseases based on symptoms entered by the user. The primary objective of the system is to assist users in obtaining a preliminary disease prediction and, in future milestones, provide disease descriptions, medications, precautions, diet plans, and workout recommendations.

In this milestone, the project focused on data understanding, exploratory data analysis (EDA), data preprocessing, machine learning model training, model evaluation, and model serialization.

2. Dataset Used

The project uses a publicly available disease-symptom dataset consisting of binary symptom values.

The project consists of the following datasets:

Dataset	**Purpose**
Training.csv	Used to train ML models
Testing.csv	Used for evaluation
symptom_Description.csv	Disease descriptions
symptom_precaution.csv	Precautions
medications.csv	Medication information
diets.csv	Diet recommendations
workout_df.csv	Workout recommendations
Symptom-severity.csv	Symptom severity scores

Dataset Characteristics

• 132 symptom features

• 41 disease classes

• Symptoms represented as binary values (0 = absent, 1 = present)

3. Environment Setup

The project was developed and executed locally using a custom virtual environment to maintain package dependencies and avoid conflicts. The implementation was done in Python 3.11.9 using Visual Studio Code as the primary IDE.

The key libraries and hardware configurations required include:

Development Environment

- Python 3.x
- Visual Studio Code
- Jupyter Notebook

Libraries Used

- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Seaborn
- Joblib

Hardware

The project was developed and executed on a Windows-based system using CPU resources.

4. Data Exploration

Before preprocessing, the datasets were explored to understand their structure and quality.

The following analyses were performed:

- Dataset shape
- Feature names
- Target classes
- Missing value detection
- Duplicate record checking
- Disease distribution
- Symptom distribution

Observations

- No significant missing values were found.
- Disease labels were balanced.
- Symptom columns were already encoded in binary format.
- Dataset was clean and suitable for supervised learning.

5. Data Preprocessing

Several preprocessing steps were performed before training the machine learning models.

5.1 Label Encoding

Disease names were converted into numerical values using LabelEncoder.

Example:

Fungal infection → 15

Malaria → 24

Allergy → 0

The trained encoder was saved as:

label_encoder.pkl

5.2 Feature Selection

Input Features:

132 symptom columns

Target Variable:

Disease

5.3 Model Serialization

The trained model and preprocessing files were saved using Joblib.

Saved files:

disease_model.pkl

label_encoder.pkl

symptoms.pkl

6. Proposed System

The proposed MedAssist AI system follows the workflow shown below.

User Symptoms

↓

Data Preprocessing

↓

Feature Encoding

↓

Machine Learning Model

↓

Disease Prediction

↓

Future Modules

↓

Description

Medicine

Diet

Workout

Precautions

7. Training Pipeline

The machine learning pipeline consists of the following stages.

Step 1

Load datasets

↓

Step 2

Perform preprocessing

↓

Step 3

Encode disease labels

↓

Step 4

Loaded training and testing data

↓

Step 5

Train six ML algorithms

- Decision Tree
- Random Forest
- Extra Trees
- Logistic Regression
- KNN
- Naive Bayes

↓

Step 6

Evaluate models

↓

Step 7

Save the best model

8. Model Evaluation

The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1 Score
- Training Time
- Prediction Time

```python
results_df = pd.DataFrame(results)

results_df = results_df.sort_values(
    by="Accuracy",
    ascending=False
)

results_df
```

0.0s

	Model	Accuracy	Precision	Recall	F1 Score	Training Time	Prediction Time
4	KNN	1.00000	1.000000	1.00000	1.00000	0.012241	0.017806
3	Logistic Regression	1.00000	1.000000	1.00000	1.00000	0.383657	0.000000
5	Naive Bayes	1.00000	1.000000	1.00000	1.00000	0.026688	0.007254
0	Decision Tree	0.97619	0.988095	0.97619	0.97619	0.090621	0.000000
1	Random Forest	0.97619	0.988095	0.97619	0.97619	0.902341	0.013586
2	Extra Trees	0.97619	0.988095	0.97619	0.97619	1.096651	0.008395

Best Model

Among all trained models,

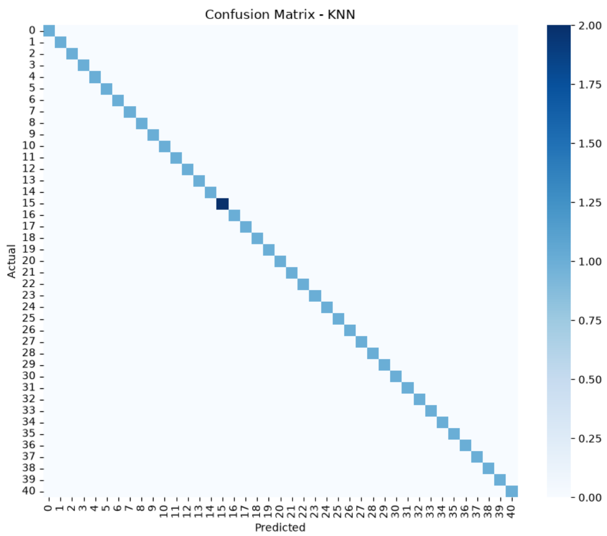
- KNN
- Logistic Regression
- Naive Bayes

achieved **100% accuracy** on the current test dataset.

Considering both accuracy and prediction speed, **KNN** was selected for deployment in the next milestone.

Confusion Matrix

The confusion matrix for the selected KNN model shows predictions concentrated along the diagonal, indicating correct classification for nearly all disease classes.



Sample Prediction

A sample from the testing dataset was used to verify the trained model.

```python
import joblib

model = joblib.load("../saved_models/disease_model.pkl")
label_encoder = joblib.load("../saved_models/label_encoder.pkl")

sample = X_test.iloc[[0]]

prediction = model.predict(sample)

predicted = label_encoder.inverse_transform(prediction)[0]
actual = label_encoder.inverse_transform([y_test[0]])[0]

print("Predicted Disease :", predicted)
print("Actual Disease    :", actual)
```

0.1s Python

```
Predicted Disease : Fungal infection
Actual Disease    : Fungal infection
```

9. Conclusion

In Milestone 1, the MedAssist AI project successfully completed data exploration, preprocessing, machine learning model training, evaluation, and model serialization. Six machine learning algorithms were compared, with KNN selected as the primary deployment model due to its high accuracy and fast prediction performance.

The trained model is now ready for integration with the backend API in the next milestone. Future work will focus on developing the backend using Flask/FastAPI, connecting the trained model with the frontend, and providing additional healthcare recommendations such as medications, diet plans, workout suggestions, and precautions.